

CTF-GS: one compact foundation-feature Gaussian map

Frozen foundation heads supervise training; the deployed scene reads the same compact map as 2D features, 3D primitives, and 3D points.

Offline supervision

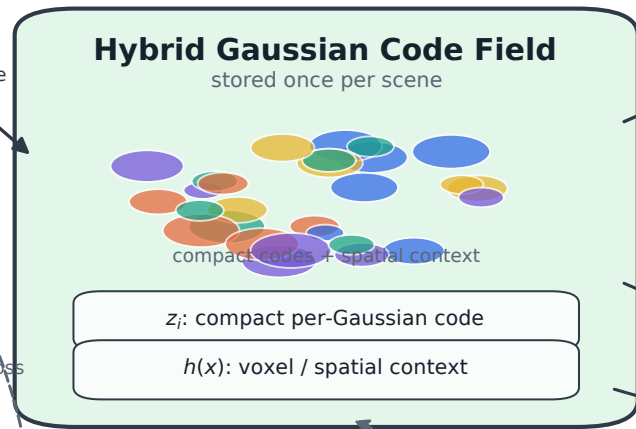
posed RGB views
+ 3DGS geometry

dense RADIO teacher
1280-D features

frozen task heads
SigLIP2 / DINO / SAM3

registered multiview
support teacher (VPR)

Stored compact map



Global readout heads

Compact-to-Teacher Decoder

align

View-Space
Aligner

Support-Aware
Primitive Policy

Frozen-Head
Adaptors

Protocol evidence

LERF rendered-view OVS
2D heatmap / mask

LERF direct 3D OVS
primitive support

ScanNet point query
open-vocabulary points

Frozen-head probes
SigLIP2 / DINO / SAM3

Training-only constraints

feature reconstruction + frozen-head consistency
visibility-aware support distillation from registered VPR
geometry regularization + SAM3 mask-logit/boundary supervision

— inference
- - - training only

Dashed paths supervise training only; deployed readouts share the stored compact map.